7

Example projects

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Outline

Problem bank → **Before you build** → **Two examples**

Problem bank

ANYONE

DOMAIN-SPECIFIC you need field knowledge to check the result

Outside your field? Read its design: the roles, the checks and where you step in.

Before you build: four things to settle

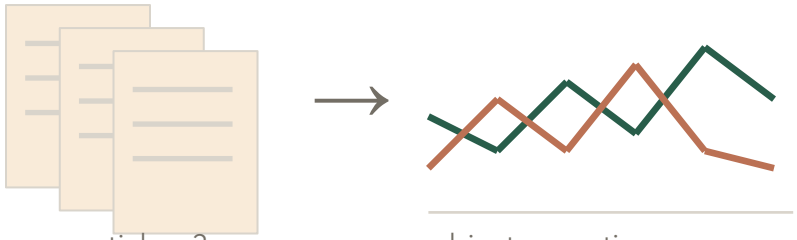
Problem — the question, in plain words

Inputs — (if any) your files, data or setup

Checks — how the verifier decides it's done;
great if there is a known answer for a case

Scope — how wide a problem the team is built for
(i.e. how detailed are agent prompts)

Example 1 of 2: news on data centers



What: A program that finds the subjects in 3 years of US news, and how each rises and falls

Why: helps a researcher pick articles to read closely

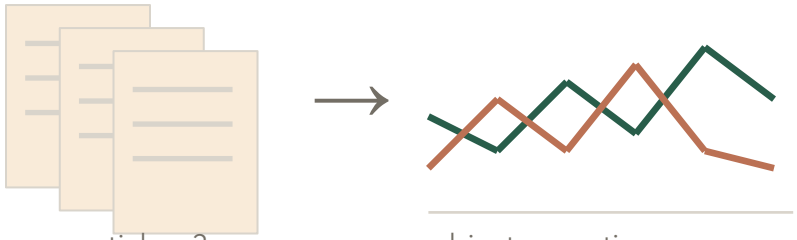
DECIDE BEFORE THE TEAM

Problem
Inputs
Checks
Scope

THE TEAM

RESULT

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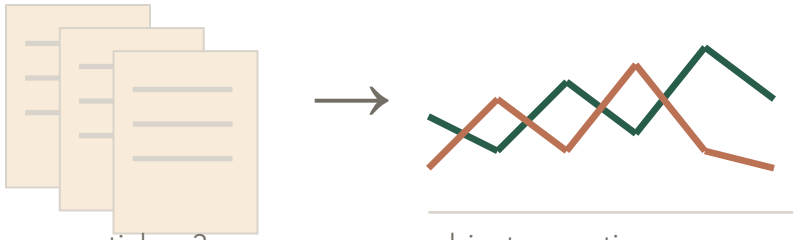
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Problem

Study how the topics of US news on data centers changed over the past 3 years, quarter by quarter.

Example 1 of 2: news on data centers



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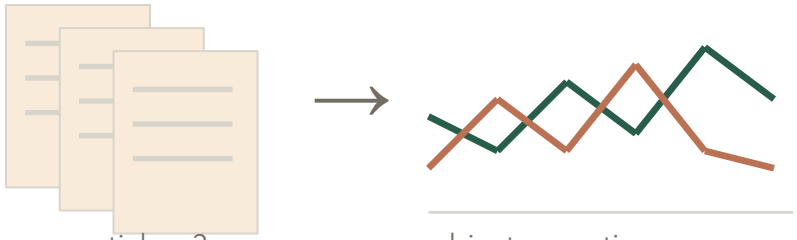
RESULT

Inputs

Oct 2023 to Sep 2026

The team finds the articles

Example 1 of 2: news on data centers



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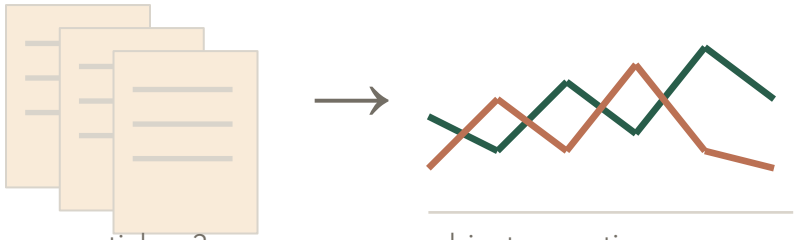
Checks

Code: same numbers on a rerun; 130+ articles, 40+ sites

Facts: correct time split; news or industry magazines

Human-in-the-loop: choose which sources count, before the search.

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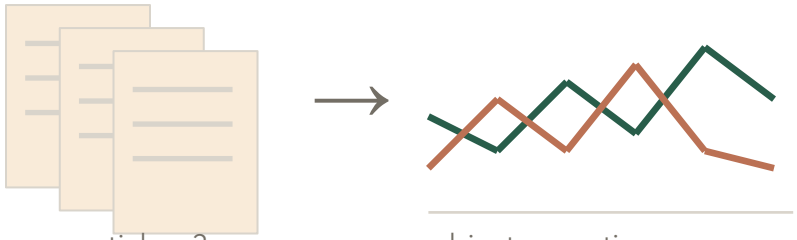
THE TEAM

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Scope

Any subject in US news, over 2 to 4 years

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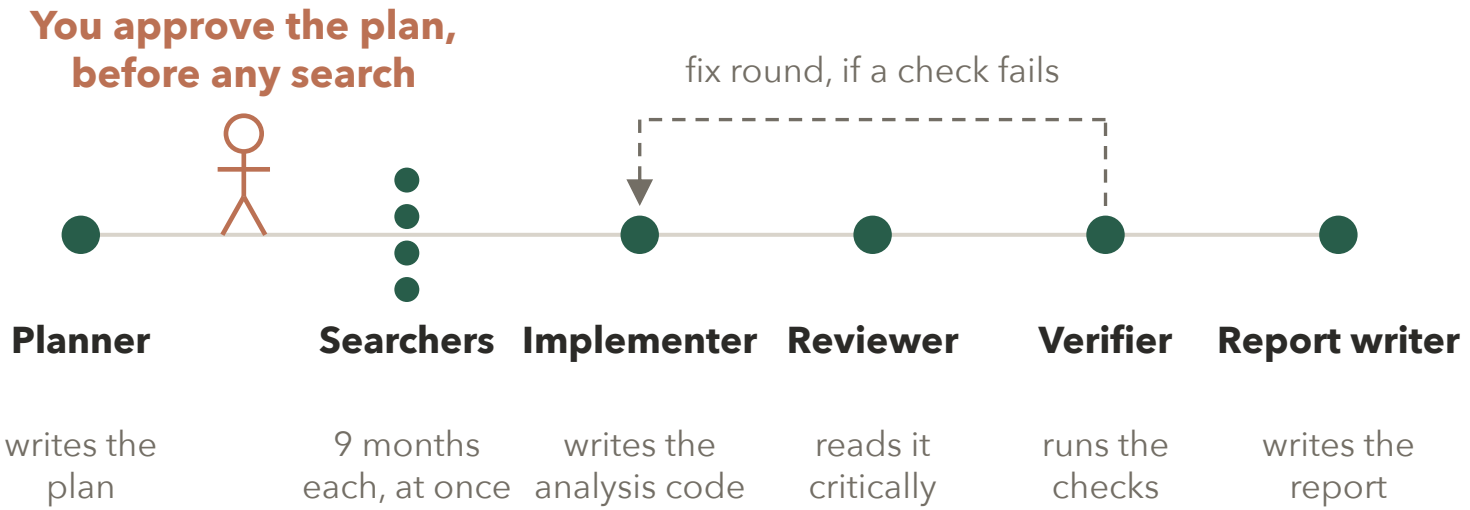
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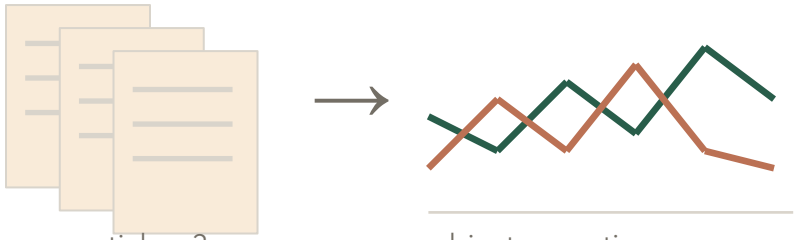
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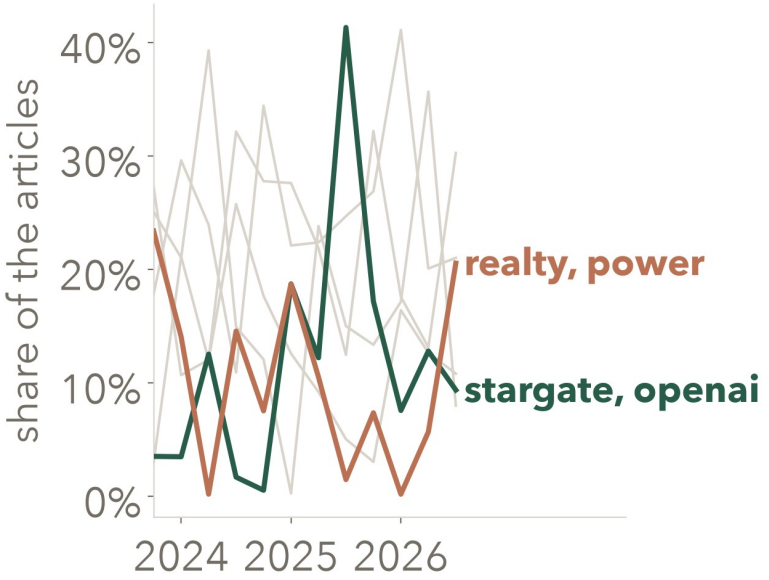
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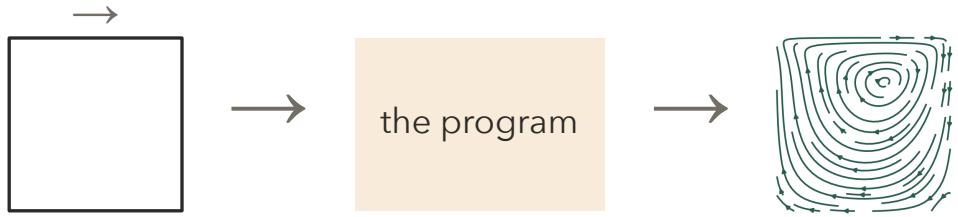
Result



About $1.50/run
Around 2 hr.

Test for smaller scope first.

Example 2 of 2: a fluid solver



a setup: the lid moves

the flow

What: A program that computes how air or water moves

Why: shows the airflow in a street canyon, the street between two tall buildings

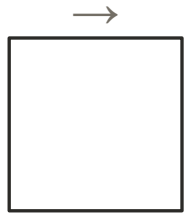
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Example 2 of 2: a fluid solver



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→ the program →



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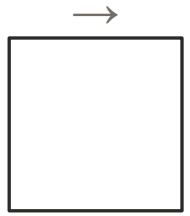
THE TEAM

RESULT

Problem

"Write a solver for slow, smooth flow in a flat rectangle, validate it, then simulate the airflow in a street 20 m wide between buildings 20 m tall. Build it for any such flow: moving walls, inflow, outflow."

Example 2 of 2: a fluid solver



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→ the program →



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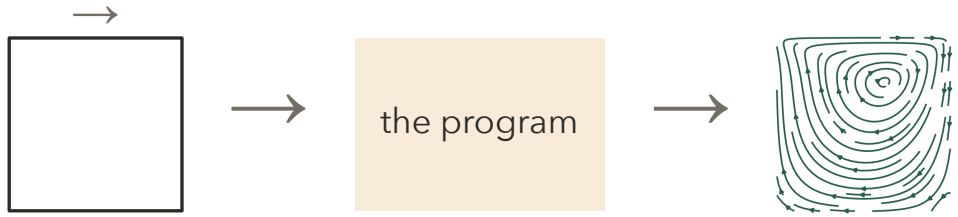
THE TEAM

RESULT

Inputs

The configuration: region size; which walls move; where fluid enters and leaves, resolution, simulation time, …

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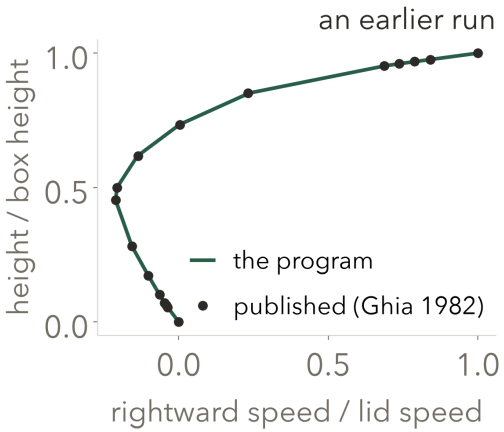
Checks

Code: it runs

Code, known answers:

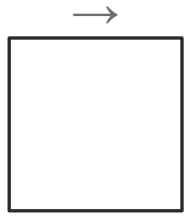
Two flat plates: exact formula, within a trillionth of the inflow's top speed. Passed

Sliding lid: published speeds at 17 points, within 5% of the lid's speed at each



Human-in-the-loop: Judge plan & validation

Example 2 of 2: a fluid solver



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the flow

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THE TEAM

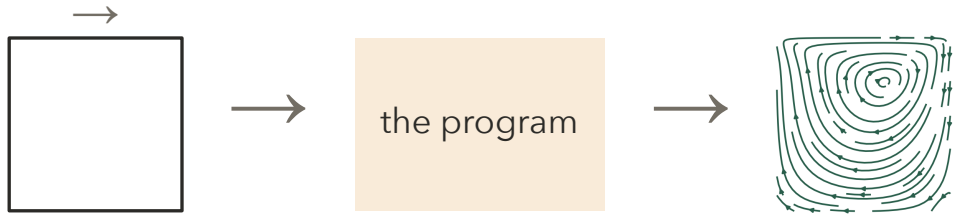
RESULT

Scope

Slow, smooth flow in a flat rectangle

Single core

Example 2 of 2: a fluid solver



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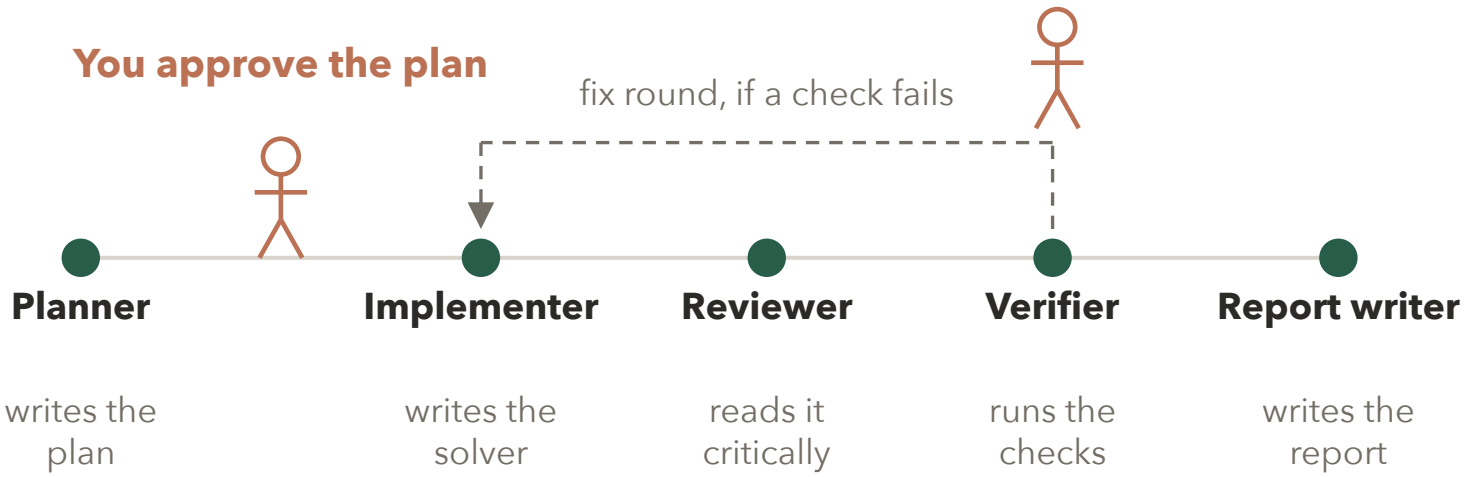
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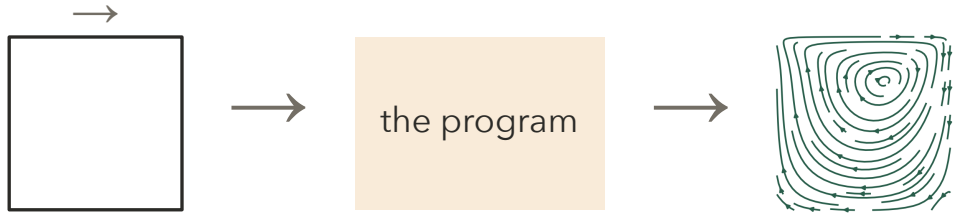
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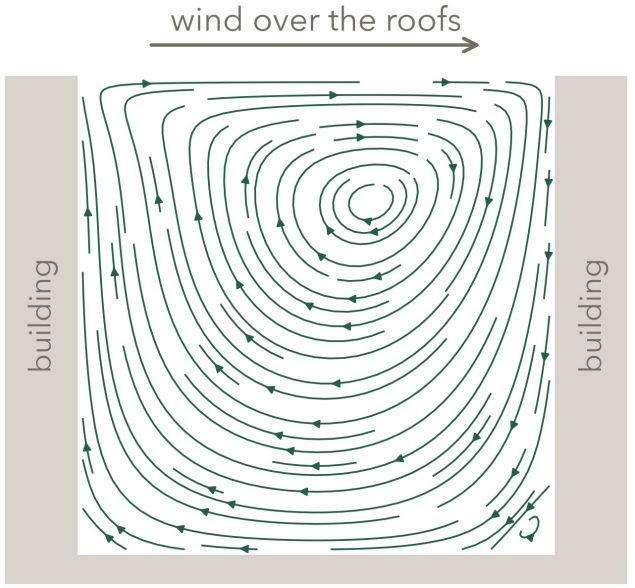
Checks

Scope

THE TEAM

RESULT

Result



A solver for any problem in scope

About $0.30/run

40 min

“

Next: walkthrough of a project & start your own